

NC28

R1A	28
	Catalytic converters catalyze redox reactions that convert gaseous pollutants in exhaust gas into less-toxic forms thereby reducing air pollution. Name one metal that is commonly employed as catalyst in automotive catalytic converters.
1	Platinum
2	Palladium
3	Rhodium

R1B	28 (30 sec)
	Give the systematic name of the major product formed in the given reaction
1	The reaction of 3-methyl cyclopentene with hydrogen gas Methyl cyclopentane (accept 1-methyl cyclopentane)
2	The reaction of 2-ethyl-1-butene with hydrogen chloride 3-chloro-3-methyl pentane
3	The reaction of 1,4-dimethyl cyclohexene with hydrogen bromide 1-bromo-1,4-dimethyl cyclohexane

R1C	28
	Assuming that the given compound exists, calculate the molar mass of element X if
1	the mass of 0.150 moles of $\text{H}_2\text{X}_2\text{O}_8$ is 26.4 g. 23.0 g/mol
2	the mass of 0.0250 moles of $\text{Ca}(\text{X}_2\text{O}_5)_2$ is 8.20 g. 32.0 g/mol
3	the mass of 0.0150 moles of $\text{NaH}(\text{X}_2\text{O}_7)_3$ is 7.11 g. 19.0 g/mol

R2	28
1	A solution was formed by dissolving 0.915 g of potassium perbromate in enough water to obtain a total volume of 200.0 cm^3. What is the molarity of this solution? [Br=80.0 g/mol] 0.0250 mol dm³ [MM=183 g/mol]
2	Give the sum of the whole number coefficients of reactants and products in the balanced equation of the reaction that occurs during the displacement reaction of lead with phosphoric acid. 9 $3 Pb + 2 H_3PO_4 \rightarrow 3 H_2 + Pb_3(PO_4)_2$
3	The reaction between a nuclide and curium-246, atomic number 96 produces nobelium-254 atomic number 102, and 5 neutrons. Identify the nuclide. Carbon-13

R4	28	
1	Caesium has the lowest first ionization energy compared with rubidium and barium	True
2	Magnesium has the smallest atomic radius compared with aluminium and scandium	False
3	Sodium has the highest electron affinity compared with lithium and calcium	False

R4	28	
1	During a sigma bond formation, p orbitals overlap head-on to form strong bonds	False
2	During hybridization in carbon, hybrid orbitals always have lower energy than the unhybridized orbitals from which they are formed	False
3	The lack of free rotation around carbon-carbon single bonds is due to the strong attraction of sigma bonds	False

R5	28
1	I am a chemical element in the news lately
2	I have properties of metals and non-metals
3	I am largely produced as a by-product of zinc production
4	I am currently the subject of an export ban from China, a country that accounts for about 60% of my supply.
5	I am named after a country
6	You will find me sandwiched between silicon and tin on the periodic table
Who am I?	
Germanium	

NC29

R1A	29
1	The oxidation state of a metal M in a nitride is +2. What is the formula of the nitride? M₃N₂
2	The oxidation state of a metal M in an oxide is +2.5. What is the formula of the oxide? M₄O₅
3	The oxidation state of a metal M in a sulfide is +5. What is the formula of the sulfide? M₂S₅

R1B	29 (30 sec)
	Give the systematic name of one isomer of the compound with molecular formula C ₅ H ₁₀ given that it is a substituted unsaturated hydrocarbon
1	2-methyl-1-butene OR 2-methyl but-1-ene
2	2-methyl-2-butene OR 2-methyl but-2-ene
3	3-methyl-1-butene OR 3-methyl but-1-ene

R1C	29
	You are given the following bond energies in kJ/mol: C≡C is 840; C=C is 614; C–C is 350; C–H is 415; C–Br is 276; and Br–Br is 193
1	calculate the magnitude of the enthalpy of bonds formed when ethyne reacts with excess bromine. 2284 kJ/mol
2	calculate the magnitude of enthalpy of bonds broken when ethene reacts with bromine. 2467 kJ/mol
3	calculate the magnitude of enthalpy of bonds formed when propene reacts with hydrogen 4020 kJ/mol

R2	29
1	What is the empirical formula of a nitrogen oxide containing 36.8% nitrogen? N₂O₃
2	Determine the mass of nitrogen dioxide formed when 0.5 mole of zinc nitrate undergoes thermal decomposition. 46 g 2 Zn(NO₃)₂ → 2 ZnO + 4 NO₂ + O₂
3	How many atoms are present in 4 formula units of 3-octanone? 100 atoms [C₈H₁₆O]

R4	29	
	For a reaction at equilibrium, increasing the concentration of reactants result in...	
1	A decrease in volume available in which particles can move without collision	True
2	An increase in the number of collisions between products per litre per second	False
3	An increase in the number of particles fulfilling the requirements of a reaction	True

R4	29	
1	Carbon-11 is a positron emitter	True
2	Sodium-25 is a negative beta emitter	True
3	Argon-42 is a positron emitter	False

R5	29
1	I'm a fundamental concept in chemical thermodynamics
2	I refer to an expression used in chemical reactions
3	My value is determined by concentration.
4	However, I don't depend on concentration
5	I help to predict the direction in which a reaction will proceed
6	At a given temperature, I remain constant
7	I describe the extent to which a reversible reaction proceeds towards products before reaching equilibrium
Who am I?	
Equilibrium constant	

R1A	30
	During the synthesis of a particular product in the laboratory, the actual yield of the product is always lower than the theoretical yield. Give <u>one</u> reason to account for this observation
1	Incomplete reactions - Not every reactant molecule reacts completely to form the desired products
2	Side reactions - Leading to the formation of unwanted products.
3	Experimental Losses – Example during transfer and purification processes
4	Impurities in starting material (Difficult to obtain 100% of each starting materials)
5	Experimental conditions - Real-world conditions may never match ideal/theoretical conditions.
6	Errors - Measurement errors, human errors

R1B	30 (30 sec)
	Give the systematic name of the major product formed when the given alkene reacts with hydrogen chloride
1	2-methyl-1-butene 2-chloro-2-methyl butane
3	1-methylcyclobutene 1-chloro-1-methyl cyclobutane
4	4-chloro-1-butene 2,4-dichlorobutane

R1C	30
1	A radioisotope has a half-life of 3.40 minutes. Given an initial amount of 38.4 g, how many grams of the radionuclide will remain after 27.2 minutes? 0.150 g
2	If it takes 11.2 hours to transport a radioisotope of half-life 1.6 hours from a laboratory to a hospital for diagnosis, what is the minimum amount of the isotope that has to be produced if 3.50 mg of it is required? 448 mg
3	After 8.96 hours, only 0.450 g of a 57.6 g sample of a radioisotope remains. What is the half-life of the radioisotope? 1.28 hours

R2	30
1	At 180 °C, methanol is formed from a reaction of carbon monoxide and hydrogen gas in an equilibrium reaction with partial pressures (atm) of 0.80, 0.40, and 2.5, respectively. Calculate the equilibrium constant of this reaction. 0.32 (atm⁻²) $K_P = \frac{0.8}{0.4 \times 2.5^2}$
2	How many moles of acid is required to convert 5 mole of iodate to iodine in a reduction reaction? 30 moles $2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightarrow \text{I}_2 + 6\text{H}_2\text{O}$
3	Calculate the sum of the average oxidation states of sulfur in sulfur dioxide, sodium thiosulphate and aluminium sulfide. +4 (+4; +2; -2)

R4	30	
1	The degree of unsaturation in butyne is 2	True
2	The degree of unsaturation in cyclopentene is 2	True
3	The degree of unsaturation in benzene is 4	True

R4	30	
1	PCl_3 is a polar molecule	True
2	NO_2 is a polar molecule	True
3	PCl_5 is a polar molecule	False

R5	30
1	I am an alcohol and a very useful one at that
2	When oxidized I form a ketone
3	I am a disinfectant and often used in wound cleaning and in hand sanitizers
4	When dilute, I am commonly referred to as “rubbing alcohol”
5	I may be prepared from the hydrogenation of acetone
6	My chemical formula is $\text{C}_3\text{H}_8\text{O}$.
Who am I?	
2-propanol OR Propan-2-ol OR isopropyl alcohol	

NC31

R1A	31
	Give the name of the element with the given symbol
1	Rg Roentgenium
2	Ts Tennessine
3	Tm Thulium

R1B	31(30 sec)
	Give the systematic name of one substituted non-cyclic structural isomer of methyl cyclobutane
1	2-methylbut-1-ene OR 2-methyl-1-butene
2	3-methylbut-1-ene OR 3-methyl-1-butene
3	2-methylbut-2-ene OR 2-methyl-2-butene

R1C	31
1	Determine the percentage composition by mass of oxygen in Butane-1,2-diol 35.5 % OR 36 %
2	Determine the percentage composition by mass of carbon in 3-methyl butanal 69.7 % OR 70 %
3	Determine the percentage composition by mass of carbon in 1-methyl cyclopentanol 72.0 % OR 72 %

R2	31
1	At a given temperature, butane effuses through a pinhole at one and a half times the rate of an unknown gas. Determine the molar mass of the unknown gas. 130.5 g/mol $\frac{1.5}{1} = \frac{\sqrt{x}}{\sqrt{58}}$
2	Determine the sum of the whole number coefficient of reactants and products in the balanced equation for the complete combustion of hexanal. 43 $2 \text{ C}_6\text{H}_{12}\text{O} + 17 \text{ O}_2 \rightarrow 12 \text{ H}_2\text{O} + 12 \text{ CO}_2$
3	Determine the mass of nitrogen present in 2.7 g of ethanamine. 0.84 g $[\frac{2.7}{45} \times 14]$

R4	31	
1	1-butene has a higher boiling point than 2-butene	False
2	2-butene has a lower boiling point than 1-pentene	True
3	2-methyl-2-butene has a lower boiling point than trans-2-pentene	True

R4	31	
1	Nitrates can easily be reduced to nitrites	True
2	Iodides can easily be reduced to iodates	False
3	Sulfates can easily be reduced to sulfides	True

R5	31
1	I am an important household chemical
2	I am a salt
3	I'm a strong oxidizing agent
4	I am formed from the reaction of chlorine with sodium hydroxide
5	I am often simply referred to as bleach
6	I have the molecular formula NaOCl
Who am I?	
Sodium hypochlorite	

NC32

R1A	32
	Give the expression for the solubility product K_{sp} for the sparingly soluble salt formed, if any, when the given compounds react
1	Silver nitrate and sodium chromate $K_{sp} = [Ag^+]^2[CrO_4^{2-}]$
2	Barium hydroxide and sodium iodate $K_{sp} = [Ba^{2+}][IO_3^-]^2$
3	Copper (II) chloride and sodium phosphate $K_{sp} = [Cu^{2+}]^3[PO_4^{3-}]^2$

R1B	32
	The given alcohol is formed from an alkene through an oxymercuration-demercuration reaction leading to the formation of a Markownikov product. Give the systematic name of the alkene.
1	3-methyl-2-butanol 3-methyl-1-butene
2	2,3-dimethyl-2-butanol 2,3-dimethyl-2-butene
3	1-ethyl cyclohexanol 1-ethyl cyclohexene

R1C	32
	Tetrabromoethene may be formed from the reaction: $6 \text{C}_2\text{H}_4\text{Br}_2 + 4 \text{Br}_2 + 5 \text{O}_2 \rightarrow 4 \text{C}_2\text{HBr}_3 + 2 \text{C}_2\text{Br}_4 + 10 \text{H}_2\text{O}$ [Br=80.0 g/mol]
1	What mass of water is formed from 5.64 g of dibromoethane given that all other reagents are in excess. 0.900 g
2	Calculate the maximum mass of tetrabromo ethene formed from 2.82 g of dibromoethane given that, all other reagents are in excess. 1.72 g
2	What is the maximum mass of tribromoethane formed if 6.88 g of tetrabromoethane is formed in a reaction? 10.6 g

R2	32
1	Given that the enthalpy of the formation of strontium oxide is -590.0 kJ/mol, determine the enthalpy change when 5.20 g of strontium oxide is formed. [Sr=88.0 g/mol]. -29.5 kJ
2	Give the sum of the whole number coefficient of reactants and products in a balanced metathesis reaction of potassium phosphate and cobalt (II) chloride. 12 $2\text{K}_3\text{PO}_4 + 3 \text{CoCl}_2 \rightarrow 6 \text{KCl} + \text{Co}_3(\text{PO}_4)_2$
3	When certain foods are cooked at high temperatures, sugars, such as glucose and fructose, can react with the free amino acid, asparagine, to form acrylamide, a potentially carcinogenic compound in humans. Determine the mass of 8.00 moles of acrylamide produced in one such reaction given that its structure is $\text{CH}_2=\text{CHCNH}_2$ 568 g

R4	32	
1	Nylon is a synthetic condensation polymer	True
2	Polystyrene is a synthetic condensation polymer	False
3	Polyethylene terephthalate is a synthetic condensation polymer	True

R4	32	
	The given reaction: $2 \text{CH}_3\text{OH} + 3 \text{O}_2 \rightarrow 2 \text{CO}_2 + 2 \text{H}_2\text{O}$ occurs under standard conditions.	
1	The enthalpy change of the reaction is the enthalpy of the formation of carbon dioxide and water	False
2	The enthalpy change of the reaction is the standard enthalpy of combustion of methanol	False
3	The enthalpy change of the reaction is the standard enthalpy of reaction of methanol	False

R5	32
1	I am a synthetic industrial chemical
2	I am a precursor to a widely used plastic
3	I am an organochloride
4	I am generally prepared by the high-temperature chlorination of ethane
5	I am commonly referred to as vinyl chloride
Who am I?	
Chloroethene (accept 1-chloroethene)	

R1A	33 (30 secs)
	Give a balanced equation for the reaction between the given compounds in an aqueous solution. You may ignore the states.
1	Sodium sulfate and ammonium iodide $\text{Na}_2\text{SO}_4(\text{aq}) + 2(\text{NH}_4)\text{I}(\text{aq}) \rightarrow 2\text{NaI}(\text{aq}) + (\text{NH}_4)_2(\text{SO}_4)(\text{aq})$
2	Lithium phosphate and barium chloride $2\text{Li}_3(\text{PO}_4)(\text{aq}) + 3\text{BaCl}_2(\text{aq}) \rightarrow 6\text{LiCl}(\text{aq}) + \text{Ba}_3(\text{PO}_4)_2(\text{s})$
3	Aluminium chloride and potassium sulphate $2\text{AlCl}_3(\text{aq}) + 3\text{K}_2\text{SO}_4(\text{aq}) \rightarrow \text{Al}_2(\text{SO}_4)_3(\text{aq}) + 6\text{KCl}(\text{aq})$

R1B	33 (30 sec)
	Give the appropriate systematic name for the given misnamed compound
1	2,4-diethyl pentane 3,5-dimethyl heptane
2	2-chloro-3-pentyne 4-chloro-2-pentyne
3	2-ethyl-2-bromo-3-butene 3-bromo-3-methyl-1-pentene

R1C	33
1	Upon heating, the temperature of an ideal gas increases from 27 °C to 87 °C, while its volume decreases from 100 cm ³ to 90 cm ³ . Determine the new pressure of the gas if its original pressure is 60 kPa. 80 kPa
2	If an ideal gas occupies a volume of 150.0 dm ³ at 327 °C and 200.0 kPa, what volume will it occupy at 137°C and 100.0 kPa? 205 dm³
3	Calculate the new temperature in degrees Celsius when an ideal gas originally occupying 40.0 m ³ at 67.0 °C and 340.0 kPa, now occupies 180.0 m ³ at 100.0 kPa. 177 °C

R2	33
1	The heating of molybdenum (IV) oxide with carbon produces molybdenum carbide and carbon monoxide. Calculate the mass of carbide formed when 2.56 g of the metal oxide reacts with an excess of carbon. [Mo = 96.0 g/mol] 2.16 g $[\text{MoO}_2 + 3\text{C} \rightarrow \text{MoC} + 2\text{CO}]$
2	Under standard temperature and pressure conditions, what is the minimum volume of oxygen gas needed to completely react with one mole of pentene? 168 dm³
3	A radioisotope has a half-life of 2.3 minutes. How long does it take for 24 g of an initial 25.6 g of the isotope to decay? 9.2 minutes $\frac{25.6}{2^4} = 25.6 - 24$, Hence 4×2.3

R4	33	
	If an organic compound decolourises bromine water, then...	
1	The compound is an alkene	False
2	The compound is an alkyne	False
3	The compound is an alkene or an alkyne	False

R4	33	
1	Decomposition reactions involve the breakdown of a compound into its constituent elements	False
2	Decomposition reactions are endothermic and absorb heat from the environment	False
3	Electrolysis is a decomposition reaction	True

R5	33
1	I'm a chemical compound with a powerful name
2	I am a yellow crystalline solid with high stability
3	I am an aromatic compound
4	Indeed, I am a substituted toluene
5	I am commonly used in explosives and often confused with dynamite
6	I am often referred to by my initials as TNT
Who am I?	
Trinitrotoluene OR 2-Methyl-1,3,5-trinitrobenzene OR 2,4,6-Trinitrotoluene	

NC34

R1A	34
	Identify the oxidizing or reducing agent in the given reaction and provide a justifiable reason for your choice.
1	Reducing agent in the reaction: $ClO_3^- + SO_3^{2-} \rightarrow Cl^- + SO_4^{2-}$ SO_3^{2-}: Oxidisation state of sulfur has increased from +4 (in SO_3^{2-}) to +6 (in SO_4^{2-})
2	Reducing agent in the reaction: $Cr_2O_7^{2-} + CH_3OH \rightarrow Cr^{3+} + CO_2$ CH_3OH: Oxidation state of carbon has increased from -2 (in CH_3OH) to +4 (in CO_2)
3	Oxidizing agent in the reaction: $Cu + HNO_3 \rightarrow Cu(NO_3)_2 + NO_2 + H_2O$ HNO_3: Oxidation state of nitrogen has decreased from +5 (in HNO_3) to +4 (in NO_2)

R1B	34
	Give the systematic name of the products formed in the given reaction, if any. Assume that the reaction takes place under gentle heating.
1	Propanoic acid and butanol react in the presence of an inorganic acid. Butyl propanoate and water
2	Pentyl methanoate reacts with dilute inorganic acid Methanoic acid and pentanol
3	Ethyl propanoate reacts with sodium hydroxide solution. Sodium propanoate and ethanol

R1C	34
	Given that Faraday's constant is 96500 C/mol
1	Calculate the current needed to deposit 0.254 g of copper from copper (I) nitrate in 77.2 seconds. [Cu = 63.5 g/mol]. 5.00 A Current = $\frac{0.254 \times 96500}{63.5 \times 77.2} = 5.00 \text{ A}$
2	Calculate the mass of gold deposited at the cathode during the electrolysis of gold (III) nitrate solution if a current of 5.00 A is applied for 193 seconds. [Au = 197 g/mol]. 5.91 g Mass = $\frac{5 \times 193 \times 3}{96500} \times 197 = 5.91 \text{ g}$
3	Calculate the volume of bromine gas produced at STP when a current of 300 mA is passed through molten potassium bromide for 8 hours. [Br = 80 g/mol] 1 dm³ vol = $\frac{0.3 \times 8 \times 60 \times 60}{96500 \times 2} \times 22.4 = 1 \text{ dm}^3$

R2	34
1	Determine the molarity of an aqueous solution of nitric acid with a density of 1.50 g cm⁻³ and mass percentage of 42.0 %. 10.0 mol dm⁻³
2	During water purification, aluminium sulfate reacts with sodium hydroxide in a metathesis reaction. Determine the sum of the whole number coefficients of reactants and products in a balanced equation of the reaction, if 2 moles of aluminium sulfate reacts. 24 $2 \text{ Al}_2(\text{SO}_4)_3 + 12\text{NaOH} \rightarrow 4\text{Al}(\text{OH})_3 + 6\text{Na}_2\text{SO}_4$
3	Give the total number of atoms present in the empirical formula then state the empirical formula itself of hex-2-en-2,4-diol. 10 atoms C₃H₆O

R4	34	
1	The partial neutralization of ammonia by hydrobromic acid results in a buffer solution	True
2	The partial neutralization of acetic acid by sodium carbonate results in a buffer solution	True
3	The partial neutralization of phosphoric acid by potassium hydroxide results in a buffer solution	True

R4	34	
	For a given reaction at equilibrium...	
1	The value of the equilibrium constants is equal to the product of the concentrations of products over that of reactants.	False
2	When the value of the equilibrium constant is large it always suggests that the reaction mostly forms products	True
3	The equilibrium constant for the reverse reaction is equal to the inverse of the equilibrium constant for the forward reaction	True

R5	34
1	I am an important chemical element in the human body
2	I am highly reactive and hardly exist naturally in the free state
3	I am commonly used in fireworks
4	I have three main allotropes named using the colours: white, red, and black
5	By far, my largest use is in fertilizers.
6	I am located in group 15 of the periodic table
Who am I?	
Phosphorus	

R1A	35
1.	The oxidation state of iron in Fe_3O_4 is +2.67. Given that electrons are neither transferred nor shared in fractions, clearly explain how this oxidation state is obtained. The 3 atoms of iron present in the compound exist in different oxidation states: (two in +3 and one in +2 state, giving a total of +8). The oxidation state is an average of these oxidation states.
2.	Iron nails left in moist air are found to have turned brownish after several days. Will the nails have the same, lower, or higher mass as before? Clearly explain. Mass is expected to be higher. Iron reacts with oxygen and moist air to form hydrated iron oxide (rust). The incorporation of oxygen and water increases the mass of the nails.
3.	A piece of dry wood is burned partially in air leaving a combination of ash, charcoal and particulate matter. Based on the law of conservation of mass, is the mass of the residual matter the same, higher, or lower than the initial mass? Clearly explain. Lower. Due to the loss of atoms as a result of the exit of carbon dioxide, carbon monoxide, volatile organic compounds and water vapour.

R1B	35
	Name the main type of organic reaction occurring in the given description
1	The formation of propane from the hydrogenation of propene Addition reaction
2	The formation of ethene from the heating of chloroethane with a base Elimination reaction
3	The formation of ethanol from the reaction of iodoethane with a base Substitution reaction

R4	35	
	For the equilibrium involving the gaseous reaction: $2 \text{NOCl} \rightleftharpoons 2 \text{NO} + \text{Cl}_2$ the equilibrium constant is 100.	
1	An increase in the volume of the reaction vessel will shift the equilibrium in favour of the products	True
2	An increase in temperature will increase the value of the equilibrium constant	False
3	The reaction is endothermic if the equilibrium constant is 78 at an increased temperature	False

R4	35	
1	Reforming largely leads to an improvement in the quantity of fuels	False
2	Cracking leads to the formation of only smaller molecular weight alkanes	False
3	Reforming always involves the use of a catalyst	True

R5	35
1	I am a white crystalline salt formed from three elements
2	I am a naturally occurring mineral commonly used in the formulation of bath salts
3	My common name is derived from a town in England where I was first discovered.
4	I may be prepared by reacting magnesium hydroxide with sulfur dioxide in air
5	I am commonly found in the hydrated form
6	My heptahydrate, also called Epsom, is the most common.
Who am I?	
Magnesium sulfate OR Magnesium tetraoxo sulfate (VI)	

R1A	36
	Identify the state of hybridization of each of the carbon atoms present in the given compound
1	1-Propanol C-1, C-2, C-3 are SP³ hybridized
2	Propene C-1 & C-2 are SP² hybridized; C-3 is sp³ hybridized
3	Propanoic acid C-1 is SP² hybridized; C-2 & C-3 are SP³ hybridized

R1B	36
	Give the systematic name of one isomeric ester of the compound with the molecular formula C ₄ H ₈ O ₂ .
1	Methyl propanoate
2	Ethyl ethanoate
3	Propyl methanoate

R1C	36
1	What is the percentage dissociation of a $2.5 \times 10^{-2} \text{ mol dm}^{-3}$ aqueous solution of a weak acid with dissociation constant 4.0×10^{-7} ? 0.40 %
2	Calculate the dissociation constant of a 0.25 mol dm^{-3} aqueous solution of a weak base that has a pH of 10. 4.0×10^{-8}
3	What mass of a strong diprotic base of molar mass 124 g/mol must be dissolved in a 250.0 cm^3 solution, to give a pH of 12.0 ? 0.155 g

R2	36
1	Determine the empirical formula of the compound given the percentage composition of elements present as 40.0% carbon, 46.6% nitrogen, 13.4% hydrogen. CH₄N
2	What mass of oxygen is present in 1.1 g of butanoic acid? 0.40 g $\left[\frac{1.1}{88} \times 32 \right]$
3	In the decay series of uranium-238, atomic number 92, what will be the atomic number of the resulting nuclide after 4 alpha decays and 2 beta decays? 86

R4	36	
1	The Solvay process is a method primarily for the production of sodium bicarbonate from brine	False
2	Ammonium chloride and carbon dioxide are byproducts of the Solvay process	True
3	Brine, limestone and ammonia are important raw materials in the Solvay process	True

R4	36	
1	Sucrose is a disaccharide of two glucose units joined through alpha linkages	False
2	Maltose is a disaccharide of two glucose units joined through alpha linkages	True
3	Lactose is a disaccharide of glucose and galactose units joined through beta linkages	True

R5	36
1	I am an alcohol.
2	I am a diol
3	I am one of the main raw material for polyester fibres.
4	I am also commonly used in making antifreeze formulations.
5	I am mainly produced from ethene.
6	I am prepared by the oxidation of ethene and the subsequent hydration of the ethene oxide formed
Who am I?	
Ethane-1,2-diol OR Ethylene glycol	

NC37

R1A	
	Pentene is to be prepared in the laboratory. Provide one <u>simple method</u> and the <u>main reactant</u> needed to achieve this synthesis.
1	Dehydration of pentanol
2	Dehydrohalogenation of chloro/bromo/iodo pentane (halo pentane)
3	Dehalogenation of vicinal dichloro/dibromo/diodo pentane (dihalo pentane)
4	Partial/controlled reduction of pentyne

R1B	37 (30 sec)
	Give the appropriate systematic name for the given misnamed compound
1	2-ethyl-4-methyl pent-2-ene 2,4-dimethyl hex-3-ene OR 2,4-dimethyl-3-hexene
2	2-chloro-3-bromo-4-ethyl pent-3-ene 3-bromo-2-chloro-4-methyl hex-3-ene OR 3-bromo-2-chloro-4-methyl-3-hexene
3	3,5-diethyl-2-methyl hex-4-ene 5-ethyl-3,6-dimethyl hept-3-ene OR 5-ethyl-3,6-dimethyl-3-heptene

R1C	37 (30 sec)
1	Calculate the solubility product of a saturated solution of Nickel (II) hydroxide given that the concentration of hydroxide ions is $1.0 \times 10^{-5} \text{ mol dm}^{-3}$ 5.0×10^{-16} (mol³ dm⁻⁹)
2	Calculate the mass of silver ions in a saturated solution of silver oxalate given that the solubility product of the salt is 4.00×10^{-12} . [Ag=108 g/mol] 0.0216 g OR 21.6 mg
3	Calculate the mass in mg of bromide ions in 250 cm ³ of a saturated solution of lead (II) bromide given that its solubility product is 4.0×10^{-15} . [Br = 80.0 g/mol] 0.40 mg

R2	37
1	Phosphorus tribromide reacts with ethanol to form bromoethane. Calculate the percentage yield of an experiment in which 0.23 g of ethanol reacts with excess phosphorus tribromide to form 0.545 g of bromoethane. 100%
2	What mass of lithium nitrate is needed to produce 3.20 g of oxygen gas by thermal decomposition? 27.6 g $4\text{LiNO}_3(\text{s}) \rightarrow 2\text{Li}_2\text{O}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
3	Assuming negligible change in volume, what mass of potassium hydroxide is needed to increase the pH of 140.0 cm ³ of a $0.0250 \text{ mol dm}^{-3}$ solution of nitric acid from 1.6 to 7.0? 0.196 g

R4	37	
1	2,3-dibromo propene exhibits geometric isomerism	False
2	2-bromo but-1-ene exhibits geometric isomerism	False
3	2-methyl but-2-ene exhibits geometric isomerism	False

R4	37	
1	In homogeneous catalysis, a catalyst made from the same element as at least one of the reactants is used	False
2	In heterogeneous catalysis, a catalyst is employed in a reaction in which two or more reactants are in different phases	False
3	In homogeneous catalysis, the catalyst employed has a stoichiometrically equivalent concentration as at least one of the reactants used in the reaction	False

R5	37
1	I am a dicarboxylic acid made by living organisms
2	I also have a hydroxy group in my structure
3	I am commonly found in vegetables and fruits, particularly apple
4	As such, my common name is derived from the Latin word for apple “malum”
5	I have 4 carbon atoms in my structure
Who am I?	
2-Hydroxybutanedioic acid OR Hydroxybutanedioic acid (Accept 2-Hydroxybutane-1,4-dioic acid OR 2-hydroxy-1,4-butanedioic acid)	

NC38

R1A	38
	Give the equilibrium constant expression for the given gaseous reaction
1	The reaction of methane with excess oxygen $K_c = \frac{[CO_2][H_2O]^2}{[CH_4][O_2]^2} \quad [CH_4 + 2O_2 \rightleftharpoons CO_2 + 2H_2O]$
2	The reaction of methane with excess chlorine gas $K_c = \frac{[CCl_4][HCl]^4}{[CH_4][Cl_2]^4} \quad [CH_4 + 4Cl_2 \rightleftharpoons CCl_4 + 4HCl]$
3	The reaction of methane with hydrogen sulfide $K_c = \frac{[CS_2][H_2]^4}{[CH_4][H_2S]^2} \quad [CH_4 + 2H_2S \rightleftharpoons CS_2 + 4H_2]$

R1B	38(30 sec)
	Give the systematic name of one stable structural isomer of methyl cyclopropene.
1	1,3-butadiene OR Buta-1,3-diene
2	1-butyne OR but-1-yne
3	2-butyne OR but-2-yne
4	Cyclobutene

R1C	38
1	The combustion of 10.2 g of a hydrocarbon produced 7.20 g of water. Determine the empirical formula of the compound. CH
2	The combustion of 10.2 g of an organic compound produced 26.4 g of carbon dioxide. Determine the percentage of carbon present in the compound. CH₅
3	The combustion of 7.00 g of a hydrocarbon produced 12.6 g of water. Determine the empirical formula of the compound. CH₃

R2	38
1	What is the minimum mass of sodium carbonate that must be heated to produce enough carbon dioxide that reacts with potassium hydroxide to form 2.76 g of potassium carbonate? 2.12 g
2	Upon evaporation of 200.0 cm ³ of a saturated solution of a sparingly soluble salt of molar mass 242 g/mol, the mass of the dry substance obtained is 9.68 mg. Determine the solubility product of the salt given that it is formed between a group 2 element and a group 17 element. 3.20×10^{-11}
3	The oxidation state of hypothetical metal M in its oxide is +2.4. Given that the atomic mass of M is 42 g/mol, what is the molar mass of the oxide? 306 g/mol [M₅O₆]

R4	38	
1	The oxidation state of chlorine in $\text{Ca}(\text{ClO})_2$ is the same as in HCl	False
2	The oxidation state of nitrogen in N_2O_5 is the same as in HNO_3	True
3	The oxidation state of potassium in $\text{H}_4\text{P}_2\text{O}_7$ is the same as in PH_3	False

R4	38	
1	Cis-2-butene has a higher boiling point than trans-2-butene and the two are enantiomers	False
2	Cis-2-butene has exactly the same density as trans-2-butene and both are disubstituted alkenes	True
3	Cis-2-butene is less thermodynamically stable than trans-2-butene and both are geometric isomers	True

R5	38
1	I'm an important concept in chemical reactions
2	I refer to a state of balance during a chemical reaction
3	I may signal the end of a chemical reaction
4	I am particularly crucial during neutralization reactions
5	You will find me on a titration curve
6	I'm the point where equivalent moles of reactants react
Who am I?	
Equivalence point (or Stoichiometric point)	

R1A	39
	Predict whether a precipitate(s) will form when the given compounds are mixed in an aqueous solution. Where a precipitate(s) is/are formed, give the formula of the precipitate(s).
1	Calcium nitrate, copper (II) nitrate, sodium chloride and lead (II) acetate PbCl₂
2	Manganese (II) nitrate, sodium sulfate, barium acetate and potassium chloride BaSO₄
3	Ammonium carbonate, iron (II) nitrate, sodium sulfate, potassium chloride FeCO₃

R1B	39 (30 sec)
	The oxidation of a branched alcohol with molecular formula C ₅ H ₁₂ O with potassium permanganate forms a carboxylic acid. Give the name of one possible structure of the alcohol.
1	2-methyl-1-butanol
2	3-methyl-1-butanol
3	2,2-dimethyl-1-propanol

R1C	39
	Atomic mass of Fe is 56.0 g/mol
1	When 1.08 g of aluminium reacts with excess aqueous iron (II) sulfate, iron is formed with a yield of 75.0%. Calculate the mass of iron formed. 2.52 g $2 \text{ Al} + 3 \text{ FeSO}_4 \rightarrow 3 \text{ Fe} + \text{Al}_2(\text{SO}_4)_3$
2	Given that, the theoretical yield of metallic iron expected in a reaction is 8.40 g, determine the highest percentage yield of metallic iron formed when a given quantity of magnetite, Fe ₃ O ₄ , is roasted in excess carbon monoxide to produce 7.04 g of carbon dioxide? 80.0% $\text{Fe}_3\text{O}_4 + 4 \text{ CO} \rightarrow 3 \text{ Fe} + 4 \text{ CO}_2$
3	Calculate the percentage yield of hydrogen gas formed given that, when 2.24 g of iron reacts with excess sulfuric acid 2.71×10^{22} molecules of hydrogen are produced under standard conditions. 75.0% $3 \text{ H}_2\text{SO}_4 + 2 \text{ Fe} \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3 \text{ H}_2$

R2	39
1	A solution containing 3.96 g of a dibasic organic acid is completely neutralised by 80.0 cm^3 of 0.55 mol dm^{-3} sodium hydroxide solution. Calculate the molar mass of the acid. 180 g/mol $[\text{H}_2\text{A} + 2\text{NaOH} \rightarrow \text{Na}_2\text{A} + 2\text{H}_2\text{O}]$
2	Cyclohexane thiol, $\text{C}_6\text{H}_{12}\text{S}$ reacts with oxygen to form carbon dioxide, water and sulfur dioxide. In a reaction, if 5.4 g of water is formed, what mass of sulfur dioxide is likely produced? 3.2 g $\text{C}_6\text{H}_{11}\text{SH} + 10 \text{O}_2 \rightarrow 6 \text{CO}_2 + 6 \text{H}_2\text{O} + \text{SO}_2$
3	A hydrated metal nitrate of molar mass 400.0 g/mol has 9 molecules of water and a metal of oxidation state +3. Identify the metal. Chromium [Molar mass of metal is 52]

R4	39	
	In the Haber process...	
1	Lowering the temperature leads to an increase in the value of the equilibrium constant of the forward reaction	True
2	Increasing the pressure and temperature of the system increases the rate of ammonia production	False
3	Iron is used as a catalyst and molybdenum is often used as a promoter	True

R4	39	
1	When HBr is added to propene in the presence of a peroxide, the major product formed is 1-bromopropane	True
2	When HCl is added to 2-methyl-2-butene, the major product formed is 2-methyl-2-chlorobutane	False
3	When HBr is added to 3-methylcyclopentene, the major product formed is 1-bromo-2-methylcyclopentane	True

R5	39
1	I am a chemical device named after my function
2	I facilitate the movement of reactants but not their mixing
3	I usually contain a pair of counter ions with similar mobility
4	You'll often find me in electrochemical cells and batteries
5	My role is to maintain electrical neutrality
6	I mediate the movement of ions between two half-cells of an electrochemical cell
Who am I?	
Salt bridge	

R1A	40
	Would the mixing of stoichiometrically equivalent quantities of the given reagents in water give rise to an acidic, basic or neutral solution? Clearly explain, indicating the formula of the product formed.
1	Ammonium hydroxide and sulfuric acid (NH₄)₂SO₄; Acidic. Ammonium ion undergoes (cationic) <u>hydrolysis to form an excess of hydroxonium ions (H₃O⁺)</u> in the solution making the solution acidic
2	Barium hydroxide and acetic acid Ba(CH₃COO)₂ or Ba(C₂H₃O₂)₂; Basic. Acetate ion undergoes (anionic) <u>hydrolysis to form an excess of hydroxide ions (OH⁻)</u> in solution making the solution basic
3	Strontium hydroxide and nitric acid Sr(NO₃)₂; Neutral. Neither strontium ion nor nitrate ion undergoes hydrolysis. Hence the solution remains neutral.

R1B	40
	An important product can be formed from the reaction of two reagents: A and B. One of these reagents is to be added in excess. Give and explain one major reason that will generally justify the selection of one reagent over the other as the reagent in excess.
1	Low cost / Less expensive / readily available – can be obtained in higher amounts
2	Low toxicity / easier to handle / environmental consideration – being in excess will not cause any health or environmental harm
3	Easier to separate from the product – The excess reagent will always be present with the product and hence ease of separation is important
4	Does not lead to side reactions/formation of unwanted products – This would lead to a low percentage purity, which is not desirable

R1C	40
	Calculate the standard enthalpy change for the formation of hydrogen chloride given... $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s}) \quad \Delta H^\circ = -175 \text{ kJ/mol}$ $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g}) \quad \Delta H^\circ = -92 \text{ kJ/mol}$ $\text{N}_2(\text{g}) + 4\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{NH}_4\text{Cl}(\text{s}) \quad \Delta H^\circ = -630 \text{ kJ/mol}$ +94 kJ/mol
2	Calculate the standard enthalpy change for the formation of ethyne given... $2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \quad \Delta H^\circ = -2600 \text{ kJ/mol}$ $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \quad \Delta H^\circ = -395 \text{ kJ/mol}$ $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l}) \quad \Delta H^\circ = -580 \text{ kJ/mol}$ +220 kJ/mol
3	Calculate the standard enthalpy change for the combustion of methane gas given... $\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \rightarrow 2\text{CH}_4(\text{g}) \quad \Delta H^\circ = -70.0 \text{ kJ/mol}$ $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \quad \Delta H^\circ = -395 \text{ kJ/mol}$ $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) \quad \Delta H^\circ = -290 \text{ kJ/mol}$ -940 kJ/mol

R2	40
1	Since July 1, 2017, the maximum level of sulfur allowed in petroleum products in Ghana is 50 mg/L, reduced from 3000 mg/L. Assuming that, petroleum in an automotive vehicle contains 25 mg/L of sulfur, calculate the maximum mass of sulfur dioxide released into the atmosphere from the burning of 15 L of this product. 0.75 g / 750 mg $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
2	When a matchstick is struck on a matchbox, potassium chlorate reacts with phosphorus, P_4 to produce tetraphosphorous decaoxide and potassium chloride. What mass of the phosphorous oxide is formed when one mole of potassium chlorate reacts? 85.2 g $10 \text{KClO}_3 + 3 \text{P}_4 \rightarrow 3 \text{P}_4\text{O}_{10} + 10 \text{KCl}$
3	The pH of a newly formulated water-soluble product of density 1.25 g/cm^3 is 3.00. This product does not meet the specification of a minimum pH of 4.00. What is the minimum volume of deionised water needed to salvage 4.00 kg of the product to meet the required specification? 28.8 dm^3 or 28800 cm^3

R4	40	
1	The percentage dissociation of acetic acid in a 5% vinegar solution is higher than in a 2% vinegar solution	False
2	Using vinegar at a temperature of 35 °C instead of 25 °C will increase the dissociation of acetic acid and pH of the solution, thereby enhancing its action.	False
3	A greater concentration of hydroxonium ions occurs in a 4% vinegar solution compared to a 3% vinegar solution and thus the 4% solution will have a lower pH	True

R4	40	
1	2-butanol is a secondary alcohol that exhibits stereoisomerism	True
2	3-methyl-3-pentanol is a tertiary alcohol that exhibits stereoisomerism	False
3	3-chloro-1-butanol is a primary alcohol that exhibits stereoisomerism	True

R5	40
1	I am a colourless organic acid with a characteristic acrid smell.
2	I may be formed from the dehydration of lactic acid.
3	I may also be produced from the partial oxidation of propene.
4	I am mainly used in the synthesis of polymers and resins which are employed in diverse applications including plastics, paints, adhesives, and polishes.
5	Perhaps, my most familiar products are nail extensions and glossy nail polishes popular among women.
6	I am the simplest unsaturated carboxylic acid and have a molecular formula $C_3H_4O_2$
Who am I?	
2-propenoic acid / Acrylic acid / Propenoic acid / Prop-2-enoic acid	